Supercompilation as Cyclic Proof Search for the Calculus of Constructions

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Abstract. Supercompilation is a powerful program optimisation technique: it can deforest intermediate data structures, fuse composed functions, and discover recursive structure automatically. However, there is no proof-theoretic foundation for supercompilation in dependently typed languages, where types depend on terms and transformations must preserve both typing and observational equivalence.
We give such a foundation by showing that supercompilation is focused cyclic sequent proof search. Driving corresponds to invertible (asynchronous) rules, case-splitting to synchronous choice rules, and folding to cyclic backlinks. A key consequence is that induction principles are not chosen upfront, but are discovered from the cycle structure of the proof graph.
We formalise this correspondence as a mechanised pipeline establishing correctness and termination, with all results proved in Rocq without axioms. This yields a principled foundation for dependently typed supercompilation, connecting program transformation with proof theory and enabling reasoning about optimisation correctness.

1 Introduction

Supercompilation is a semantics-driven program optimisation technique that can automatically deforest intermediate data structures, fuse composed functions, and discover recursive schemes. It works by symbolically evaluating programs (driving), analyzing cases on unknown values (splitting), and recognising repeated configurations (folding). Despite decades of development for first-order and simply-typed languages, supercompilation lacks a proof-theoretic foundation for *dependently typed* languages. In such languages, types depend on terms, contexts evolve during evaluation, and transformations must preserve both typing and observational equivalence.

This paper proposes such a foundation: *supercompilation is focused cyclic sequent proof search*. Driving steps are invertible (asynchronous) sequent rules. Case-splitting on neutral scrutinees is a synchronous choice rule. Memoisation (folding) corresponds to cyclic backlinks that close proof goals by referencing previously-seen vertices. The trace condition that ensures soundness of cyclic proofs matches the termination discipline that justifies folding.

A central insight is that *induction principles are discovered post-hoc from the cycle structure*: rather than choosing an induction scheme at the outset of a proof, the cycle formed by backlinks *is* the induction principle, read off from the completed proof graph. This is particularly valuable in dependent types,

where type indices containing computations (e.g., Vec A (length (map f v))) must be normalised before rewriting, and the induction scheme must simultaneously simplify the index and prove the property.

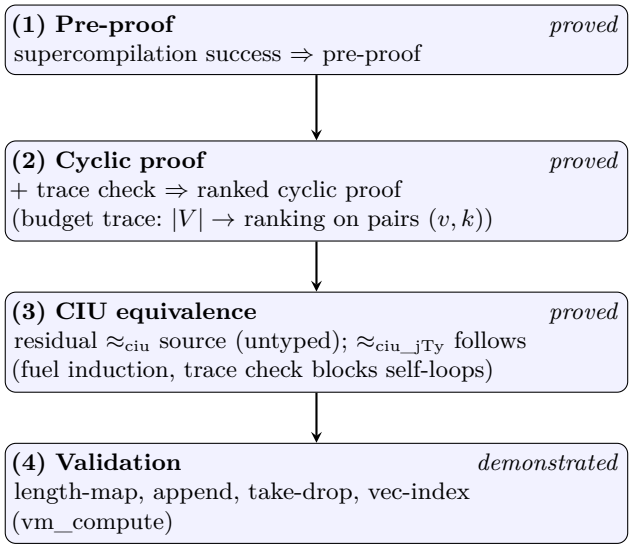
This correspondence is not merely a representation change—it has explanatory power. The trace condition explains *why* folding is sound (every cycle contains a progress event); the cyclic structure explains *where* recursion comes from (backlinks are discovered during search); and the CIU equivalence theorem explains *what* correctness means for the residual program (indistinguishable from the source under all observations).

Contributions.

- A focused cyclic sequent calculus designed to mirror supercompilation operations (Section 4).
- A precise correspondence between drive/split/fold/generalise and local sequent inference steps, mechanised in Rocq (Section 8).
- A four-claim pipeline mechanised with zero axioms: pre-proof construction, budget trace cyclic proof, CIU soundness, and example validation (Section 8).
- A proof-theoretic account of post-hoc induction discovery via cyclic proofs, with a dependent-type motivation (Section 1.3).

Outline. Section 1.1 states the four-claim pipeline. Sections 3–4 define the source calculus and the focused cyclic sequent system. Section 5 presents the operational dictionary. Section 7 discusses related work, and Section 8 gives the formal results. The mechanisation is described in the appendix.

1.1 Main result pipeline



The four claims form a dependency chain: each tier builds on the previous. All are mechanized in Rocq with full proofs (0 axioms, 0 admitted lemmas).

Claim 1: Supercompilation yields a pre-proof. When the supercompiler succeeds on a well-typed input, the resulting configuration graph directly forms a *rooted pre-proof*. Every vertex satisfies its local sequent rule and each edge is justified by the appropriate correspondence (Section 2). This claim is mechanized in Rocq; see Section 8 (Theorem 1).

Claim 2: Termination yields a cyclic proof. A pre-proof becomes a *cyclic proof* when it additionally satisfies a global progress condition: every cycle must contain at least one progress edge (a case-split on a neutral scrutinee), and there exists a well-founded ranking that strictly decreases along progress edges. The supercompiler enforces this via a trace-condition check during graph construction. We prove this by constructing a budget-instrumented trace graph: vertices are lifted to pairs (v, k) where $k \in [0, B]$ is a budget counter consumed on progress edges and preserved on non-progress edges. The resulting trace graph is acyclic, yielding a full ranking_condition with rank $\lambda(v, k).\, k$. This claim is mechanized; see Section 8 (Theorem 2).

Claim 3: CIU equivalence. The ultimate soundness guarantee is *contextual equivalence* (CIU): the program read off from a valid cyclic proof should be observationally indistinguishable from the original source program under all closing substitutions. This claim is mechanized by fuel induction; the trace condition prevents self-loops that would introduce non-termination. See Section 8 (Theorem 3).

Claim 4: The pipeline obtains for our examples. The framework is validated on a suite of standard supercompilation examples (length-map fusion, append length normalisation, take-drop, etc.). For each example, the supercompiler produces a residual, and computational equality with the expected result is verified by Rocq's `vm_compute`. See Section 8 (Proposition 2) and Section 6 for a detailed walkthrough.

1.2 Why focusing matters

Focusing separates proof search into an asynchronous (invertible) phase and a synchronous (choice-bearing) phase. Driving is deterministic (async); generalisation and folding are the choice points (sync); case-splitting sits at the boundary. See Section 2 for the correspondence table.

1.3 Why cyclic proofs matter: post-hoc induction

In traditional induction, the scheme must be chosen upfront. Cyclic proofs reverse this: begin with the goal, drive and split following the term structure, and form backlinks when goals match previous vertices. The cycle structure *is* the induction principle, read off from the completed graph. The trace condition ensures soundness via structural descent through progress events.

This is particularly valuable in dependent types. Consider $\mathsf{Vec}\ A\ (\mathsf{length}\ (\mathsf{map}\ f\ v))$: the type index contains a computation that must be normalised before rewriting. A traditional proof must choose an induction scheme that simultaneously normalises the index and proves the property. Cyclic proofs avoid this: driving normalises the index during search, and when a fold-back is discovered, the index equality is already reduced to definitional form. Our mechanised examples include vector-index normalisation where supercompilation reduces the index automatically.

2 Overview: SC = cyclic proof search

Before presenting formal definitions, we sketch the core idea informally.

A supercompiler maintains a *configuration graph* where each node is a typing judgement $\Gamma \vdash t : A$ and each edge represents an operational step. This graph *is* a proof object in a cyclic sequent calculus:

SC concept	Proof concept	Rule type
Driving (unfolding, β, ι)	Invertible sequent rule	Async (deterministic)
Case-split on neutral variable	Synchronous choice rule	Sync (progress event)
Memo table hit (folding)	Cyclic backlink	Cycle closure
Generalisation + instantiation	Substitution lemma	Async (invertible)

Cyclic proofs require a **global trace condition** for soundness: every cycle must contain a *progress event*—a case-split on a neutral scrutinee. This matches the termination discipline used in supercompilation: a recursive call must descend on a structurally smaller component. The budget trace construction (Claim 2) enforces this by assigning a decreasing rank to each progress edge.

A particularly striking consequence is that **induction principles are discovered post-hoc**. In traditional proofs, you must choose an induction scheme at the start. In cyclic proofs, the backlink itself *is* the induction hypothesis, formed when search recognises that the current goal is an instance of a previous one. The cycle structure *is* the induction principle, read off from the completed proof graph. This matches how supercompilation works: the residual program's recursive structure is discovered by the driving process, not prescribed in advance.

The remainder of the paper formalises this intuition: Section 3 defines the term calculus, Section 4 presents the sequent rules, Section 5 establishes the correspondence, and Section 8 states the mechanised theorems.

3 Term calculus

The source language is the Calculus of Constructions extended with inductive types and fixpoints. Terms follow the grammar:

$$\begin{aligned} t, A, B ::= {} & x \mid \mathsf{Sort}\, i \mid \Pi(A).\, B \mid \lambda(A).\, t \mid t\, u \\ \mid {} & \mathsf{fix}(A).\, t \mid \mathsf{Ind}^I(\boldsymbol{a}) \mid \mathsf{roll}^I_c(\boldsymbol{a}) \mid \mathsf{case}^I(t; C; \boldsymbol{b}) \end{aligned}$$

where x ranges over de Bruijn indices, $\mathsf{Sort}\, i$ is the universe hierarchy, $\Pi(A).\, B$ is the dependent function type, $\lambda(A).\, t$ is lambda abstraction, $t\, u$ is application, $\mathsf{fix}(A).\, t$ is a recursive definition (the recursive call is bound at de Bruijn index 0), $\mathsf{Ind}^I(\boldsymbol{a})$ is an applied inductive type, $\mathsf{roll}^I_c(\boldsymbol{a})$ is a constructor application, and $\mathsf{case}^I(t; C; \boldsymbol{b})$ is dependent pattern matching with motive C. Contexts Γ are lists of types; the global environment Σ records inductive definitions.

The typing judgement $\Sigma; \Gamma \vdash t : A$ is standard for CoC with inductives [5,11]. It includes rules for variables, universe cumulativity, dependent functions, fixpoints, inductive type formation and introduction, and dependent elimination. Conversion uses β-convertibility.

Operational semantics. We use call-by-name (CBN) weak-head reduction with three rules:

$$(\lambda(A).\, t)\, u \to t[u/] \qquad \mathsf{fix}(A).\, t \to t[\mathsf{fix}(A).\, t/] \qquad \mathsf{case}^I((\mathsf{roll}^I_c(\boldsymbol{a})); C; \boldsymbol{b}) \to b_c\, \boldsymbol{a}$$

plus congruence for reduction in the function position of application and the scrutinee of case expressions. Under CBN, arguments are never evaluated before substitution; reduction does not proceed under binders. The reflexive-transitive closure is written $\to^*$.

Observational equivalence. We use CIU (Closed Instantiations of Uses) equivalence: $t \approx u$ iff for all closing substitutions σ and CBN values v, $t[\sigma]$ terminates to v exactly when $u[\sigma]$ does. This captures the standard notion of contextual equivalence for CBN and is the correctness criterion for our supercompilation pipeline (Claim 3).

4 Focused cyclic sequent calculus

A sequent has the form $\Gamma \vdash t : A$, representing the goal of finding a term t of type A in context Γ. A *cyclic proof* is a finite directed graph where each vertex v is labelled with a sequent, has successors $\mathsf{succ}(v)$ (its premises), and satisfies a local rule. The graph may contain cycles. A proof is valid if every vertex's local rule holds and every infinite path contains infinitely many *progress events* (splits).

4.1 Asynchronous rules (driving)

The asynchronous rules are deterministic and invertible. They correspond to CBN reduction: β-reduction, fixpoint unfolding, ι-reduction (case-of-constructor), and commuting conversions. All have the same shape—a single premise that is the result of one computation step—and are applied eagerly without backtracking.

4.2 Synchronous rule: case split

The synchronous rule introduces choice by analyzing a neutral variable:

$$\frac{\Gamma \vdash b_0 : C[\mathsf{roll}_0^I()/] \qquad \cdots \qquad \Gamma \vdash b_{n-1} : C[\mathsf{roll}_{n-1}^I()/]}{\Gamma, x : \mathsf{Ind}^I(\boldsymbol{a}) \vdash \mathsf{case}^I(x; C; \boldsymbol{b}) : C[x/]} \quad (\textsc{split-case-var})$$

Each premise corresponds to one constructor of the inductive type. **Key property:** this is a *progress event*. Every cycle must pass through at least one split, ensuring structural descent.

4.3 Backlink rules (folding)

A backlink closes a goal by pointing to a previous vertex. This is the cyclic aspect: rather than choosing an induction scheme upfront, the backlink *is* the induction hypothesis, discovered when a current goal is recognized as an instance of a previous one. The trace condition ensures the resulting recursion is well-founded.

$$\frac{}{\Gamma \vdash t : A} \quad (\textsc{back-exact}) \qquad \frac{\Delta \vdash \sigma :_s \Gamma}{\Gamma \vdash t[\sigma/] : A[\sigma/]} \quad (\textsc{back-inst})$$

BACK-EXACT applies when a prior vertex has a syntactically identical label. BACK-INST applies when a prior vertex matches under an explicit substitution σ. The judgement $\Delta \vdash \sigma :_s \Gamma$ states that σ is an explicit substitution from context Γ to Δ, i.e., a list of terms in Δ that can replace the variables of Γ.

4.4 Phases and focusing

Rules are partitioned into two phases: an **asynchronous** phase (driving, deterministic) and a **synchronous** phase (split and backlink, requiring choice). This matches supercompilation: drive eagerly until stuck, then split or fold.

4.5 Global progress condition

A cyclic proof is sound if every infinite path contains infinitely many progress events (splits). A *trace state* tracks the inductive variable being analyzed; a well-founded order requires strict decrease on at least one back-edge per cycle. This is enforced by the budget trace construction (Claim 2).

4.6 Vertex classification

Each vertex in the configuration graph is classified by its successor structure:

Successors	Rule	SC operation	Phase
$n = 0$	DR-LEAF	leaf (axiom)	—
$n = 1$, $\mathsf{drive\_cbn\_once}(t) \neq t$	DR-CBN-ONCE	driving (async)	invertible
$n = 1$, $\mathsf{drive\_cbn\_once}(t) = t$	DR-FOLD	folding (memo hit)	cyclic closure
$n > 1$	DR-SPLIT	case-split (sync)	choice point

Proposition 1 (Vertex classification). *Every vertex in the configuration graph produced by successful supercompilation satisfies exactly one of the four rules above.*

5 Supercompilation as focused proof search

We give the intended dictionary between standard supercompilation concepts and focused cyclic proof search.

5.1 Configurations as sequents

A supercompiler configuration (term + context + type/motive) is represented as a sequent goal node. The supercompiler's configuration graph is then a proof-search graph.

5.2 Driving = invertible sequent steps

Driving decomposes the goal by deterministic computation-like rules (β, iota, commuting conversions). In focusing, these correspond to invertible/asynchronous rule applications.

5.3 Case splitting = left rules on neutral scrutinees

When the scrutinee is neutral, supercompilation propagates information by splitting into one successor per constructor. In a sequent system this is a left rule whose premises correspond to constructor cases.

5.4 Folding/memoisation = cyclic closure

Folding corresponds to closing a goal by back-linking to a previously seen companion sequent, recording the instantiation (explicit substitution arguments).

5.5 Generalisation

Generalisation introduces a more general sequent that subsumes the current goal and a previous goal, enabling folding. In the current implementation, a `best_generalize` heuristic selects a companion from the memo table and computes a generalised config via anti-unification. The anti-unification produces a generalised judgement S_{gen} together with explicit substitutions σ, σ_0 such that $S = \sigma(S_{\mathsf{gen}})$ and $S_0 = \sigma_0(S_{\mathsf{gen}})$. The stream-based oracle model described in the conclusion generalises this: the heuristic can be replaced by an external oracle producing candidates validated by deterministic kernel checks.

5.6 Concrete correspondence examples

We illustrate the correspondence using two representative cases: asynchronous driving (deterministic computation) and synchronous branching with folding (the source of recursion and cyclic structure). Other cases follow the same pattern.

Example 1: β-reduction (asynchronous driving)

$$\text{Supercompiler: } \Gamma \vdash (\lambda x.\, t)\ u : B \ \rightsquigarrow\ \Gamma \vdash t[u/x] : B$$

$$\text{Sequent: } \quad \frac{\Gamma \vdash t[u/x] : B}{\Gamma \vdash (\lambda x.\, t)\ u : B}$$

Both represent a single deterministic computation step. The supercompiler produces one successor configuration, and the proof graph has a single premise. All asynchronous rules (fixpoint unfolding, iota reduction, commuting conversions) follow this same pattern.

Example 2: Split + fold (synchronous + cyclic)

$$\text{Supercompiler: } \begin{array}{l} \Gamma, l : \mathsf{List} \vdash \mathbf{case}\ l\ \mathbf{of}\ \{nil \to b_0; cons \to b_1\} : A \\ \rightsquigarrow \Gamma \vdash b_0 : A \\ \quad\ \Gamma, x, xs \vdash b_1[(cons\ x\ xs)/l] : A \end{array}$$

$$\text{Sequent: } \quad \frac{\Gamma \vdash b_0 : A \qquad \Gamma, x, xs \vdash b_1[(cons\ x\ xs)/l] : A}{\Gamma, l : \mathsf{List} \vdash \mathbf{case}\ l\ \mathbf{of}\ \{\ldots\} : A}$$

This is the only source of nondeterminism in both systems.

In the recursive branch, further driving yields a goal of the form:

$$\Gamma, xs \vdash \mathsf{length}\,(\mathsf{map}\ f\ xs)$$

This matches a previously encountered goal under substitution $l \mapsto xs$. The supercompiler performs a memo hit (fold), and the proof graph closes via a cyclic backlink.

This creates a cycle in both graphs. The cycle represents the recursive structure: the induction principle is discovered from the cycle itself. The trace condition ensures the cycle is well-founded by requiring that it passes through a progress point (the case split).

These examples illustrate the general pattern: supercompilation constructs a graph whose structure coincides with a focused cyclic proof, with asynchronous steps for computation, synchronous branching at case splits, and cyclic closure via folding.

6 Concrete Example: Length-Map Fusion

To illustrate the correspondence between supercompilation and cyclic proof search, we present a worked example: the classic deforestation problem of fusing `length` and `map`.

6.1 The input term

Consider the composed function that applies a function to every element of a list, then computes the length:

$$\lambda f.\, \lambda l.\, \texttt{length}\,(\texttt{map}\ f\ l)$$

This creates an intermediate list that is immediately consumed, which is inefficient. The expected optimized output is simply:

$$\lambda f.\, \lambda l.\, \texttt{length}\, l$$

6.2 Supercompilation trace

The supercompiler proceeds as follows:

1. **Initial configuration:** $\Gamma \vdash \lambda f.\, \lambda l.\, \texttt{length}\,(\texttt{map}\, f\, l) : (\texttt{Nat} \to \texttt{Nat}) \to \texttt{List} \to \texttt{Nat}$
2. **Drive (unfold):** Unfold `length` and `map` fixpoint definitions, exposing:
$$\lambda f.\, \lambda l.\, \texttt{case}\, l\, \texttt{of}\, \{\, \texttt{nil} \to \cdots \mid \texttt{cons}\, x\, xs \to \cdots \}$$
3. **Split (case analysis):** Split on the neutral variable l, creating two branches:
   - **Nil branch:** $\texttt{length}\,(\texttt{map}\, f\, \texttt{nil}) \rightsquigarrow 0$
   - **Cons branch:** $\texttt{length}\,(\texttt{map}\, f\,(\texttt{cons}\, x\, xs))$
4. **Drive (beta-iota):** In the cons branch, drive performs:
$$\texttt{length}\,(\texttt{cons}\,(f\, x)\,(\texttt{map}\, f\, xs)) \rightsquigarrow \texttt{succ}\,(\texttt{length}\,(\texttt{map}\, f\, xs))$$
5. **Fold (memoization):** Recognize that $\texttt{length}\,(\texttt{map}\, f\, xs)$ is a recursive call matching the initial configuration. Create a backlink to the root.
6. **Result:** The residual term is:
$$\lambda f.\, \lambda l.\, \texttt{case}\, l\, \texttt{of}\, \{\, \texttt{nil} \to 0 \mid \texttt{cons}\, x\, xs \to \texttt{succ}\,(\texttt{REC}\, f\, xs)\, \}$$
where `REC` is the recursive call.

Note that the intermediate list construction `map f l` never appears in the residual term—it has been fused away.

6.3 Corresponding cyclic proof

The supercompilation trace above corresponds to a cyclic sequent proof with the following structure:

1. **Root node:** Sequent $\Gamma \vdash \lambda f.\, \lambda l.\, \texttt{length}\,(\texttt{map}\, f\, l) : (\texttt{Nat} \to \texttt{Nat}) \to \texttt{List} \to \texttt{Nat}$
2. **Asynchronous edges:** From unfolding `length` and `map`, each drive step creates a single successor with the driven term.
3. **Synchronous branching:** The case split on l creates two premises (nil and cons branches).
4. **Cyclic backlink:** The fold creates an edge back to the root node, forming a cycle.
5. **Local validity:** Each node satisfies its sequent rule:
   - Driving steps satisfy `dr_cbn_once`
   - The case split satisfies `dr_split_case_var`
   - The backlink is justified by configuration equality

6.4 Graph structure

The resulting proof graph has three nodes: the root (initial configuration), a nil branch returning 0, and a cons branch performing a recursive call. The cons branch has a backlink to the root, forming the cycle.

The cycle (from the cons branch back to root) represents the recursive structure. The *trace condition* for this cycle is satisfied because the case-split is a *progress point*: the recursive call is on xs, which is structurally smaller than l.

This example is mechanised in Rocq; the supercompiler successfully processes the input and the step correspondence theorems ensure a matching cyclic sequent proof. Additional benchmarks (append-length, append-associativity via 2-pass supercompilation, take-drop fusion, and vector-index normalisation) are validated by computational reflection.

7 Related work

This section positions the present work relative to (i) supercompilation and program transformation, (ii) cyclic proof theory, (iii) sequent calculi for dependent types, and (iv) proof-theoretic treatments of induction.

7.1 Supercompilation

Supercompilation is a program transformation technique based on symbolic evaluation (driving), case analysis, generalization, and fold/refold steps [16,13]. The key operations are:

- **Driving**: unfolding function definitions and performing symbolic reduction
- **Splitting**: case analysis on unknown values (neutrals)
- **Generalization**: abstracting common patterns to enable folding
- **Folding**: recognizing repeated configurations and creating recursive calls

Termination control. A central challenge in supercompilation is ensuring termination. Classical approaches use *whistles* based on homeomorphic embedding or well-quasi-orderings [8,9,14]. These approaches analyze configuration histories and force generalization when a dangerous pattern is detected. Our cyclic proof framework replaces the whistle with a *global trace condition*: we allow arbitrary folding but require that every cycle passes through a progress event (a split on a structurally smaller inductive component).

Supercompilation and partial evaluation. Early work connects supercompilation to partial evaluation [7]. Our contribution extends this connection to dependent types: we show that supercompilation *is* proof search in a typed setting, with configurations as sequents and the memo table as a cyclic proof graph.

Prior work on dependent types. To our knowledge, this is the first formulation of supercompilation for a dependently typed core language. Existing work on supercompilation focuses on first-order functional languages (Refal, Scheme, Haskell subsets) or simple typed settings [13,7]. The dependent type context introduces new challenges:

- Types depend on terms, so driving must respect typing constraints
- The context Γ changes during proof search (extended in branches)
- Observational equivalence is more subtle (CIU for dependent types)

Our sequent calculus formulation addresses these challenges by making the typing context explicit in every sequent and using type-directed driving rules.

7.2 Cyclic proof theory

Cyclic proof systems represent potentially infinite derivations using finite directed graphs with back-edges, validated by a global soundness condition [3,4].

Concrete cyclic systems. Brotherston and Simpson develop cyclic proofs for first-order logic with inductive definitions, using a global trace condition to ensure soundness. Their work establishes that cyclic proofs are as expressive as standard proofs with explicit induction rules, but offer computational advantages (the induction scheme emerges from the cycle structure).

Abstract cyclic frameworks. Afshari and Wehr develop a modern, highly abstract account of cyclic proof checking based on trace conditions, activation algebras, and categorical structure [1]. Our work instantiates their abstract framework in a concrete setting: CoC with inductives, CBN operational semantics, and observational equivalence.

Cyclic proofs as normal forms. Our contribution extends cyclic proof theory in a new direction: we use cyclic proofs not just as proof objects, but as *normal forms for program transformations*. The correspondence with supercompilation shows that cyclic proofs are stable under driving, splitting, and folding—operations that would be difficult to justify in a traditional natural deduction setting.

7.3 Sequent calculi for dependent types

Bidirectional type checking. Bidirectional type systems [12] separate synthesis (inferring types) from checking (validating against known types). This discipline is standard in dependently typed proof assistants (Agda, Rocq, Lean). Our sequent calculus uses a similar decomposition, but with a crucial difference: we use *focused* sequents that separate asynchronous (deterministic) from synchronous (choice-bearing) phases.

Focused sequent calculi. Focusing [2,10] is a proof search discipline that reduces nondeterminism by eagerly applying invertible rules (asynchronous phase) and carefully controlling choice points (synchronous phase). Focusing has been applied to linear logic, classical logic, and intuitionistic logic, but **to our knowledge this is the first focused sequent calculus for CoC with inductives**.

The key novelty is the connection to supercompilation: the asynchronous phase corresponds to driving (deterministic unfolding), and the synchronous phase corresponds to splitting and folding (the essential choice points).

Sequent calculi for CIC. Standard presentations of CIC use natural deduction [5,11]. Herbelin develops a sequent calculus for the Calculus of Constructions [6], but without inductive types or cyclic backlinks. Our calculus extends Herbelin's work by adding:

- Inductive types and pattern matching
- Cyclic backlinks (folding)
- Focused phases (asynchronous vs synchronous)
- Connection to supercompilation

7.4 Induction: local rules vs global discharge

Traditional induction principles. In standard type theory, induction is a derived principle from the elimination rule for inductive types. To prove a property $P(x)$ for all $x : \mathsf{List}$, one applies the list eliminator (recursor) with a motive P and proofs for the base and step cases. This requires *choosing the induction principle upfront*—a significant practical burden.

Cyclic induction as post-hoc discovery. Sprenger and Dam compare local well-founded induction rules to cyclic systems with global discharge conditions [15]. They show these approaches are equivalent in expressive power but differ operationally: cyclic proofs discover the induction scheme during proof search rather than requiring it upfront.

Our work extends this insight to dependent types and connects it to supercompilation:

- The backlink corresponds to recognizing a recursive configuration
- The trace condition corresponds to structural descent (termination)
- The cycle structure *is* the induction principle, read off from the proof graph

This flexibility is essential for advanced optimizations (fusion, deforestation, tupling) where the appropriate induction principle is not obvious from the problem statement.

8 Main results

We organise the central results as a four-claim pipeline, each building on the previous. All results are fully mechanised in Rocq; a mapping from theorems to artefact names and file paths is provided as supplementary material.

8.1 Claim 1: Pre-proof correspondence

Theorem 1 (Pre-proof). *Successful supercompilation yields a rooted pre-proof whose edge structure corresponds to the drive, split, and fold operations.*

The three step-level correspondences—driving, splitting, folding—are proved at the graph level via a bisimulation.

8.2 Claim 2: Cyclic proof via budget trace

Theorem 2 (Cyclic proof). *With the trace-condition check enabled, supercompilation yields a ranked cyclic proof on a budget-instrumented trace graph.*

The boolean trace check guarantees cycle-progress. A budget construction lifts the base graph to pairs (v, k) where progress edges consume budget. This yields a full ranking condition.

8.3 Claim 3: CIU soundness

Theorem 3 (CIU soundness). *The residualised program is CIU-equivalent to the source: for every closing substitution σ and CBN value v, $t[\sigma]$ terminates to v iff the residual does. The typed variant follows directly.*

The proof uses fuel induction with the trace check preventing non-terminating self-loops; component lemmas lift CIU through all term constructors (appendix).

8.4 Claim 4: Example validation

Proposition 2 (Examples). *Length-map fusion, append-length normalisation, append-associativity (2-pass), take-drop fusion, and vector-index normalisation are all validated by computational reflection.*

9 Conclusion

This paper develops a focused cyclic sequent proof-search view of supercompilation for a dependently typed calculus (CoC with inductives). The central claim is operational: supercompilation moves (drive/split/fold) coincide with local inference steps in a cyclic proof graph, and the cyclic global progress condition matches the structural-descent termination discipline that justifies folding.

Contributions. We organise the results as a four-claim pipeline, each mechanized in Rocq:

1. **Pre-proof construction (proved):** Supercompilation success always yields a locally-valid rooted pre-proof whose edge structure corresponds exactly to drive/split/fold operations.

2. **Cyclic proof via budget trace (proved):** With the trace-condition check enabled, the pre-proof upgrades to a ranked cyclic proof on a budget-instrumented trace graph.
3. **CIU soundness (proved):** The residualised program is CIU-equivalent to the source. The proof uses fuel induction with the trace condition preventing non-terminating self-loops. Component lemmas lift CIU through all term constructors.
4. **Example validation (demonstrated):** Concrete supercompilation examples (length-map fusion, append length, take-drop, etc.) pass the pipeline with computational equivalence verified by Rocq's `vm_compute`.

Limitations. The current presentation focuses on a sequent system tailored to the supercompilation correspondence rather than a full CIC sequent calculus; several CIC-specific features (conversion, universe cumulativity, dependent elimination obligations) are deferred.

Future work. Immediate next steps are: (i) extending the focused calculus to cover a fuller CIC sequent system (conversion, universe cumulativity, dependent elimination), and (ii) strengthening the generalisation heuristic (whistle) to optimise more complex patterns such as nested deforestation. We also view oracle-guided generalisation as promising future work. The current heuristic can be replaced by an external oracle proposing candidate substitutions, validated by the same deterministic kernel checks. The oracle is not trusted for soundness.

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A Mechanization

The four-claim pipeline is mechanized in Rocq 9.1.0 using the `stdpp` library for finite maps and graphs. The development totals ∼5,000 lines of proof across 14 files. All theorems below are proved with no axioms or admitted lemmas.

A.1 Claim 1: Pre-proof correspondence

Theorem 4 (Pre-proof construction). *For any environment Σ, fuel f, context Γ, term t, and type A, if* `supercompile_jTy` $f\ \Sigma\ \Gamma\ t\ A = \mathsf{Some}\,(v, scb)$, *then scb directly forms a* rooted_preproof *with root v.*

Coq: `all_supercompiled_programs_yield_preproof` (`SupercompilationCorrespondence.v:1659`).

Theorem 5 (Step correspondence). *Each supercompilation edge corresponds to a sequent rule.*
1. ***Driving (async):*** *`drive_corresponds_to_async_edge` (line 465)*
2. ***Splitting (sync):*** *`split_corresponds_to_sync_edge` (line 513)*
3. ***Folding (backlink):*** *`memo_corresponds_to_fold` (line 581)*

Proposition 3 (Vertex rule classification). *Every cfg_builder vertex satisfies a specific drive rule determined by its successor structure: 0 successors ⇒ leaf; 1 successor with driving ⇒ async; 1 successor without change ⇒ fold; $n > 1$ successors ⇒ split.*
Coq: `cfg_vertex_drive_rule` (`SupercompilationCorrespondence.v`).

A.2 Claim 2: Cyclic proof via budget trace

Theorem 6 (Cyclic proof from trace check). *If the supercompiler succeeds with the trace-condition check,* `supercompile_jTy_tc` $f\ \Sigma\ \Gamma\ t\ A = \mathsf{Some}\,(v, scb)$, *then there exists a* Ranked.rooted_cyclic_proof *on a budget-instrumented trace graph with budget $B = |\mathsf{dom}(scb.\mathsf{cb\_label})|$ and rank $\lambda(v, k).\, k$ ordered by $<_{\mathbb{N}}$.*

Coq: `supercompile_yields_rooted_cyclic_proof` (`SupercompilationCorrespondence.v`).

The construction (∼140 lines) instantiates the general budget-trace theorem from `CyclicTraceConditionBudget.v`:

1. The base graph is cfg_graph(scb), with progress predicate `is_progress_vertex`.
2. The trace graph lifts each base vertex v to pairs (v, k) for $0 \leq k \leq B$. Progress edges consume one unit of budget; non-progress edges preserve it.
3. Pigeonhole argument: cycle-progress implies the trace graph is acyclic, yielding a full ranking_condition.
4. Labelling each trace vertex with the judgement of its base vertex (permissive rule) gives a preproof on the trace graph.

A.3 Claim 3: CIU soundness

Theorem 7 (CIU soundness, untyped). *If* `supercompile_jTy_tc` *succeeds on t, the residualised term is CIU-equivalent to t:*

$$\forall \sigma, v.\ \texttt{terminates\_to}\ t[\sigma]\ v \iff \texttt{terminates\_to}\ t_{res}[\sigma]\ v$$

Coq: `supercompile_ciu_soundness_untyped` (`SupercompilationCorrespondence.v`).

The proof is by induction on the fuel parameter of the residualiser, using the following component lemmas (all proved):

- `step_ciu`, `steps_ciu`: any CBN step preserves CIU (`Equiv/CIU.v`)
- `ciu_tApp`, `ciu_apps`, `ciu_shift`, `ciu_subst0`, `ciu_case_branches_ciu`: CIU lifts through all term constructors (`Equiv/CIU.v`)
- `terminates_to_tApp_decompose`, `terminates_to_tCase_decompose`: CBN decomposition lemmas (`Semantics/Cbn.v`)
- `drive_cbn_once_ciu`, `whnf_drive_ciu`: supercompiler driving preserves CIU (`SupercompilationCorrespondence.v`)
- `ciu_fix`, `ciu_fix_nonrec`: fixpoint unrolling preserves CIU (`Equiv/CIU.v`)
- `trace_condition_ok_no_self_loop`: the trace check blocks cycles without progress (`SupercompileTraceCheckSound.v`)

The typed variant `supercompile_ciu_soundness_typed` follows in 8 lines: the untyped CIU quantifies over all substitutions, and the typed restriction instantiates the list-based substitution of `ciu_jTy`.

Theorem 8 (Typing preservation for leaf vertices). *For a leaf vertex (no successors) labelled Σ; $\Gamma \vdash t : A$ and adequate fuel, the residual has type A in context Γ.*
Coq: `residualise_cfg_root_typing` (`SupercompilationCorrespondence.v`).

The typing proof uses `has_type_subst` (renaming preserves typing, all 9 rules), `has_type_weaken_head` (weakening by one binder), and `has_type_shift` (uniform shift preserves typing) — all proved in `Judgement/Typing.v`.

A.4 Claim 4: Example validation

The example suite in `SupercompileChecklistIndexPipeline.v` covers length-map fusion, append-length normalisation, append associativity (2-pass), take-drop

fusion, map-map composition, and vector-index normalisation. All equalities are verified by computational reflection (`vm_compute`) with no axioms.

A.5 Summary of Rocq artefacts

Component	Key lemmas	File
Syntax & semantics	term, CBN evaluation	`Syntax/Term.v`, `Semantics/Cbn.v`
Typing	`has_type`, `has_type_subst`	`Judgement/Typing.v`
Graph theory	`fin_digraph`, `ranking_condition`	`Graph/FiniteDigraph.v`, `Progress/Ranking.v`
Pre-proofs & cyclic proofs	`preproof`, `cyclic_proof`	`Preproof/Preproof.v`, `CyclicProof/`
Supercompiler	`supercompile_cfg`, `residualise_cfg`	`Transform/Supercompile.v`
Trace check soundness	`trace_condition_ok_cycle_progress`	`SupercompileTraceCheckSound.v`
Budget trace ranking	`budget_trace_ranking_condition`	`CyclicTraceConditionBudget.v`

All pipeline theorems: `SupercompilationCorrespondence.v`

Claim 1	`all_supercompiled_programs_yield_preproof`, `drive/split/fold_corresponds_to_edge`, `cfg_vertex_drive_rule`	
Claim 2	`supercompile_yields_rooted_cyclic_proof`	
Claim 3	`supercompile_ciu_soundness_untyped/typed`, `residualise_cfg_root_typing`	
Claim 4	`CIUChecklistLengthMap`, `SupercompileChecklist-IndexPipeline`	`Equiv/` `Equiv/`

All proofs are closed (`Qed`); the development contains zero axioms or admitted lemmas.